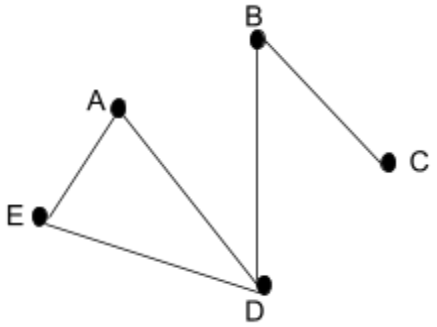
American Computer Science League

2021-2022 ● Contest 4: Short Problem Solutions ● Elementary Division

1. Graph Theory

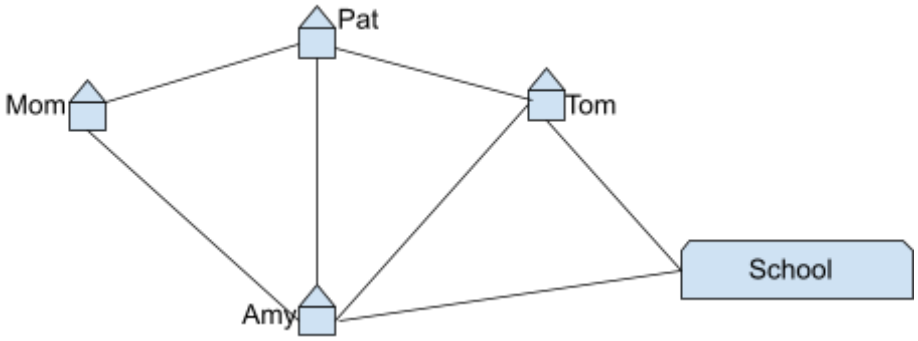
In the following graph, only removing BD will create 2 unconnected graphs with vertices {B,C} and {A,D,E}.



D. BD

2. Graph Theory

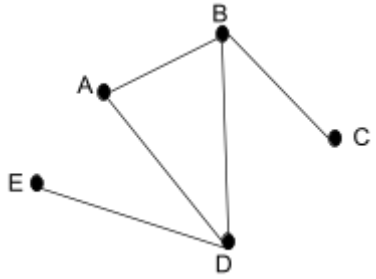
By inspection, the possible ways are MPATS, MPTAS, and MAPTS.



B. 3

3. Graph Theory

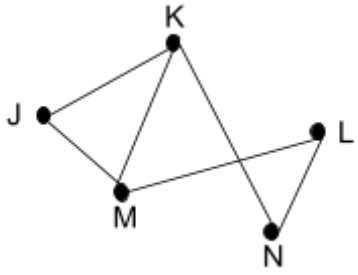
By inspection, the possible paths between 2 friends are: BAD, DAB, ABC, ABD, DBC, DBA, CBA, CBD, ADB, ADE, EDA, BDA, BDE, EDB. Ben and Deb are both the middle person 6 times. C and E can never be in the middle. Ann is only the middle person twice.



D. Ben, Deb

4. Graph Theory

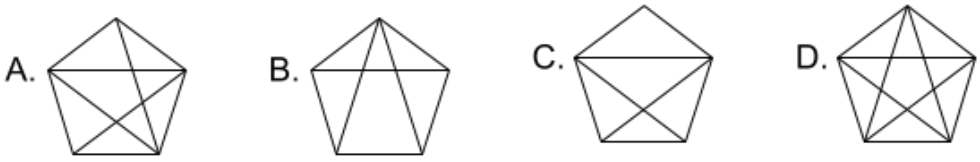
By inspection the cycles in the following graph are: JKMJ, KNLMK, and KJMLNK. By counting them both ways, there are 6 of them.



C. 6

5. Graph Theory

In the following graphs, A is traversable using the top and bottom left vertices, and C is traversable using the bottom left and bottom right vertices. In D, none are of odd degree so it is traversable regardless of which vertex is used for the start and the end. In B, there are 4 vertices with odd degree so it is not traversable.



A. A, C, D